

CONCEPT PRIMER

AI for *engineers.*

The vocabulary, mental model, tooling habits, and safety rules students need before coding with AI.

LLMs

agents

prompting

context

secrets

START HERE

The words people throw around.

AI

Systems that perform tasks associated with intelligence: language, reasoning, vision, prediction.

ML

Systems trained from examples instead of hand-written rules.

MODEL

The trained system that takes input and produces output.

WEIGHTS

Numbers inside the model that encode learned patterns.

LLM

A model trained on massive text and code datasets to generate language.

PROMPT

The input you give the model: instructions, code, errors, and context.

LLM BASICS

Models read and write tokens in context.

TOKEN

A word piece, word, or punctuation mark. The model reads and writes tokens, not whole paragraphs at once.

Token limits decide how much code, conversation, and documentation the model can use.

CONTEXT WINDOW

The total text the model can see right now: your prompt, chat history, attached files, and tool results.

missing context makes the model guess.

TRAINING VS USE

Training makes it. Inference uses it.

TRAINING

- ✓ Teaches a model from massive datasets.
- ✓ Happens before you use that model version.
- ✓ Expensive, slow, and compute-heavy.
- ✓ Creates the learned weights.

INFERENCE

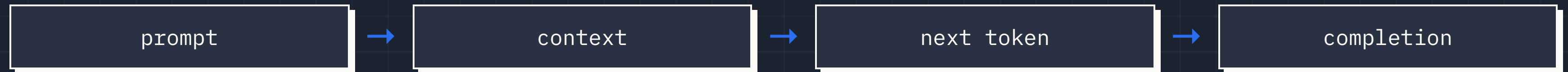
- ✓ Runs when you send a prompt.
- ✓ Uses the trained model to generate output.
- ✓ Fast enough for chats and tools.
- ✓ Still limited by available context.

knowledge cutoff means: not in training unless you provide it or a tool fetches it.

SIMPLIFIED TRUTH

An LLM predicts the next token.

Given everything before it, the model predicts likely next tokens. Repeat that enough times and you get a paragraph, a function, or a plan.



powerful pattern matching, not human understanding.

STRENGTHS AND LIMITS

Great first draft. Not automatic truth.

USEFUL FOR

- ✓ Boilerplate code.
- ✓ Explaining concepts several ways.
- ✓ Summarizing long text.
- ✓ Translating between languages and APIs.
- ✓ Drafting tests, docs, and refactors.

RISKY WHEN

- ✗ It lacks your exact code.
- ✗ The library changed recently.
- ✗ The task has hidden constraints.
- ✗ It invents a function or package.
- ✗ You accept output without running it.

read it, verify it, test it.

CONFIDENT AND WRONG

Hallucination is made-up certainty.

ENGINEERING EXAMPLES

- ✗ A package name that does not exist.
- ✗ A method not in the library API.
- ✗ A CLI flag from an old version.
- ✗ A plausible explanation that misses the real bug.

REDUCE IT

- ✓ Paste the exact error.
- ✓ Provide the real code.
- ✓ Include package versions.
- ✓ Run the output.
- ✓ Read the diff before trusting it.

CHATBOT VS AGENT

Agents use tools and iterate.

A chatbot answers and stops. An agent takes a goal, gathers context, edits files, runs commands, observes results, and keeps going.

Agent loop:

Think → Use tool → Observe → Think again → ...until done

TOOL USE

The tools change what AI can do.

FILES Read existing code and write changes.	TERMINAL Run tests, linters, installs, and scripts.	WEB Look up current docs and changing facts.
DATABASE Query data when connected safely.	APIS Call services and inspect responses.	EDITOR Explain, autocomplete, and refactor in place.

Claude Code, Codex, and Cursor Composer are agents, not just answer boxes.

GLOSS LEVEL

Terms you will hear in the wild.

EMBEDDINGS

Numbers that represent text or images so software can measure similarity.

RAG

Retrieve relevant documents, put them in context, then generate an answer.

FINE-TUNING

Train a base model further on a specific dataset.

MULTIMODAL

Models that handle text, images, audio, or video.

AI API

Services expose models through APIs. Developers pay per token.

SYSTEM PROMPT

Hidden instructions that set behavior before the user prompt.

BETTER INPUTS

Good prompts reduce guesswork.

A GOOD PROMPT INCLUDES

- ✓ Context: what you are working on.
- ✓ Goal: what output you want.
- ✓ Constraints: what to avoid or preserve.
- ✓ Format: how the answer should be shaped.

Weak

Fix my code.

Better

I'm building a React form using controlled inputs.
When I submit, the page refreshes before state updates.
Here is the component: [paste code].
Explain the cause, then show the minimal fix.
Do not rewrite the whole file.

USE THE RIGHT TOOL

Own the work. Protect secrets.

Tool	Type	Best for
Claude Code	Terminal agent	Repo-wide tasks, debugging, multi-file edits
Codex CLI	Terminal agent	Coding tasks, explanations, implementations
Cursor	Editor AI	In-file edits, codebase Q&A, completions
Warp	AI terminal	Command help, explaining output, shell workflows

never paste API keys, .env contents, passwords, or tokens into AI tools.